

Experiment No: 6

Title: Installing OS on a Virtual Machine Monitor.

Theory:

VMware ESXi step-by-step Installation

VMware ESXi is free. However, the software comes with a 60 days evaluation mode. You should register on VMware website to get your free license key to come out of the evaluation mode. Once the ESXi is installed, you can either use vSphere Client or the Direct Console User Interface to administer the host.

VMware ESXi is based on hypervisor architecture that runs directly on top of hardware as shown below.



1. Download ESXi server

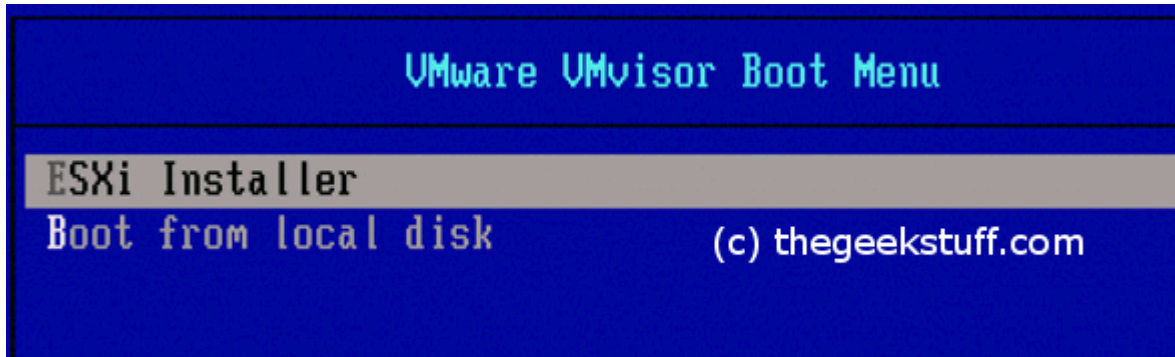
Get the software from the VMware ESXi download page.

Following are the various download options available. Select “ESXi 4.0 Update 1 Installable (CD ISO) Binary (.iso)” and burn a CD.

- ESXi 4.0 Update 1 Installable (CD ISO)
- Upgrade package from ESXi Server 3.5 to ESXi Server 4.0 Update 1
- Upgrade package from ESXi Server 4.0 to ESXi Server 4.0 Update 1
- VMware vSphere Client and Host Update Utility

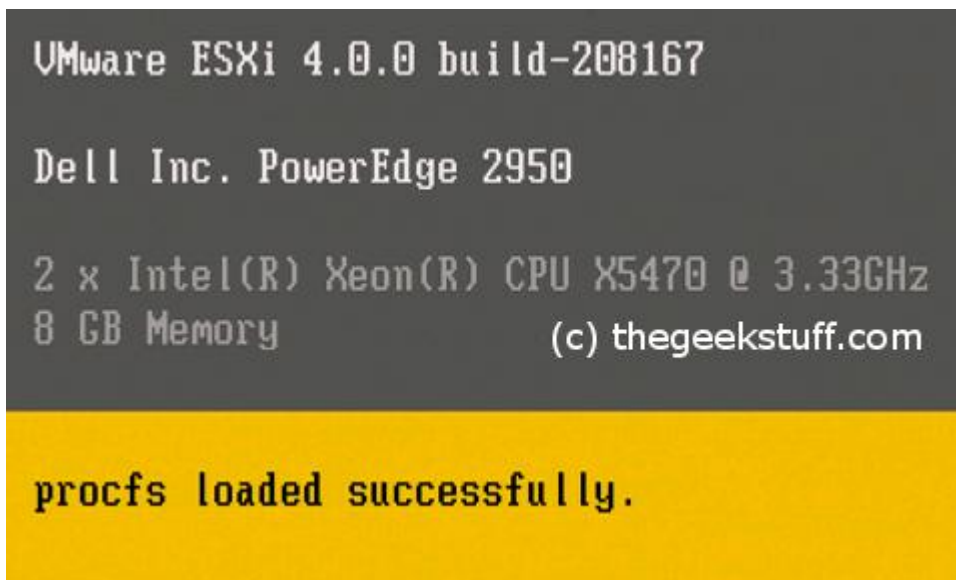
2. VMware VMvisor Boot Menu

Once you insert the ESXi CD and reboot the server, it will display a boot menu with an option to launch “ESXi Installer” as shown below.



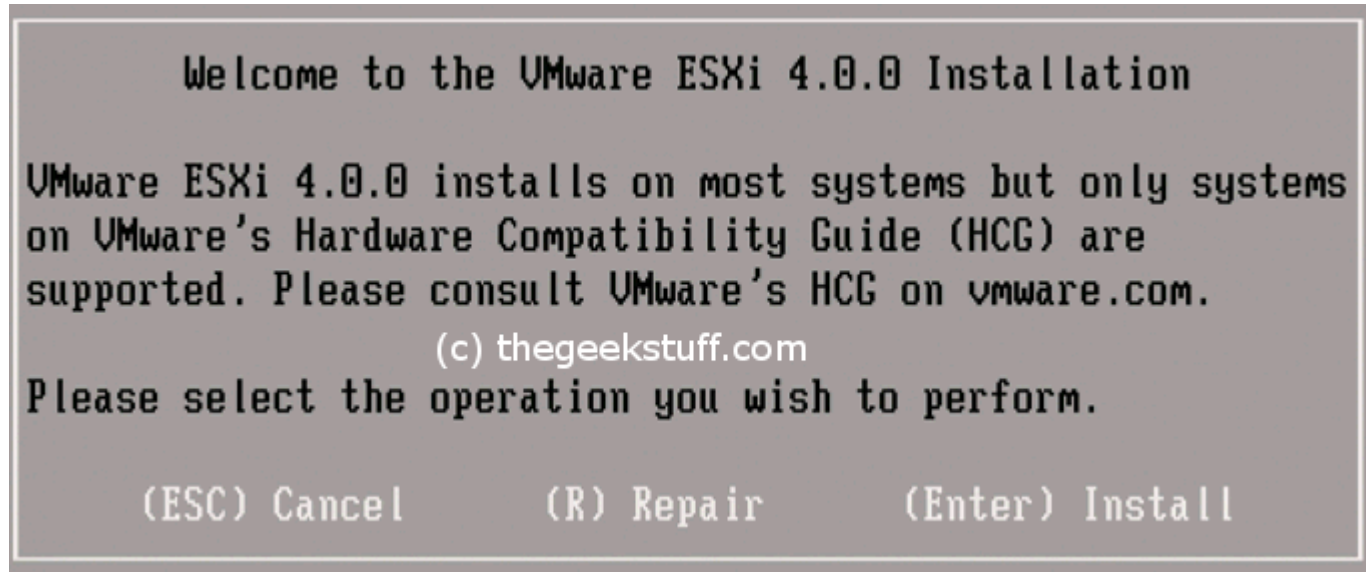
3. VMware ESXi Installer Loading

While the installer is loading all the necessary modules, it will display the server configuration information at the top as shown below. In this example, I was installing VMware ESXi 4.0 on a Dell PowerEdge 2950 server.



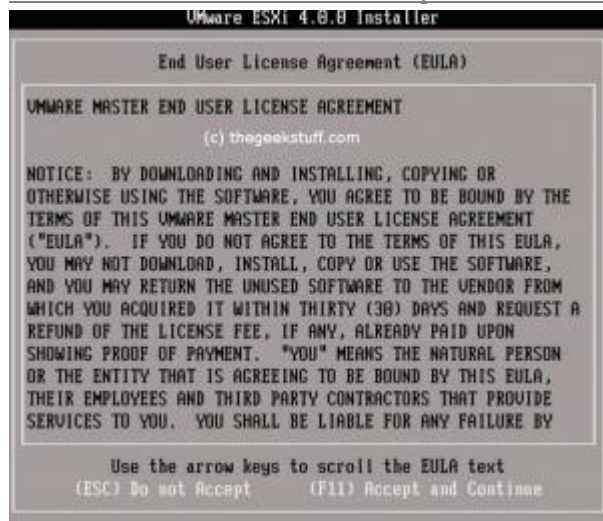
4. New ESXi Install

Since this is a new installation of ESXi, select "Install" in the following screen.



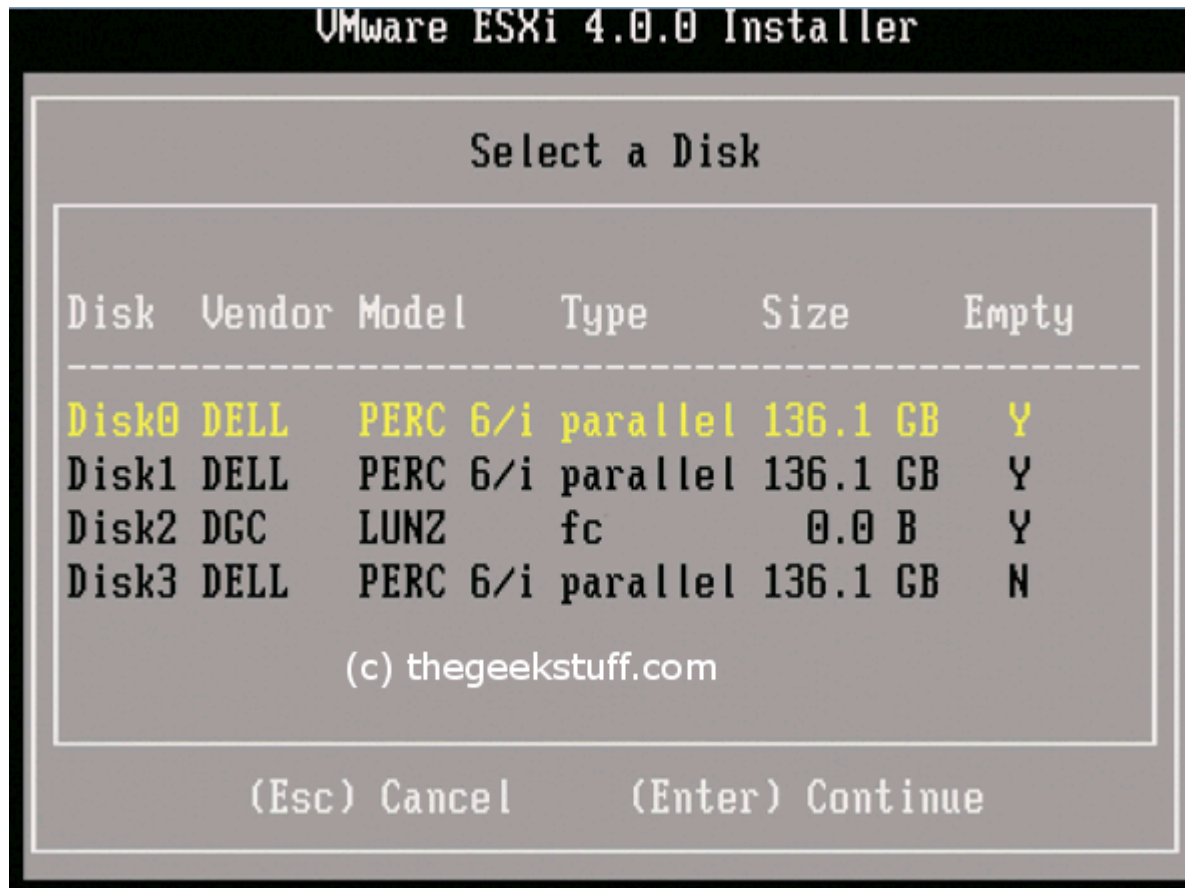
5. Accept VMware EULA

Read and accept the EULA by pressing F11.



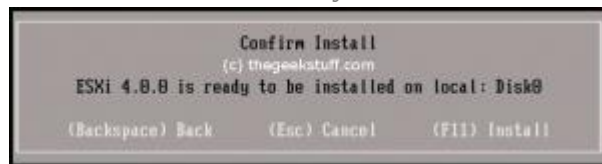
6. Select a Disk to Install VMware ESXi

VMware ESXi 4.0.0 Installer will display all available disk groups. Choose the Disk where you would like to install the ESXi. It is recommended to choose the Disk0.



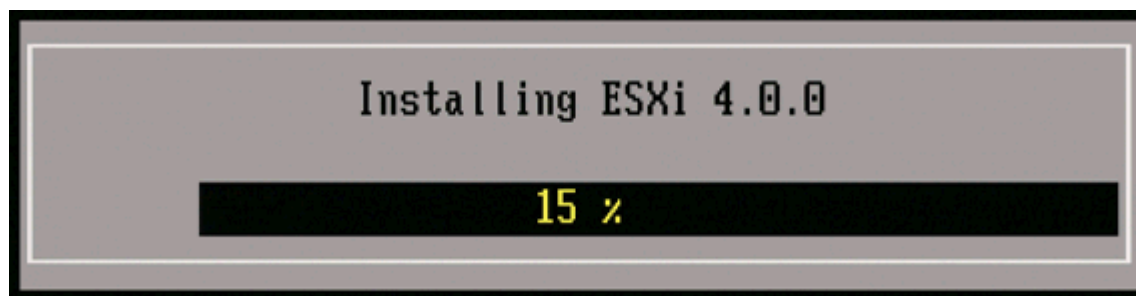
7. Confirm ESXi Installation

Confirm that you are ready to start the install process.



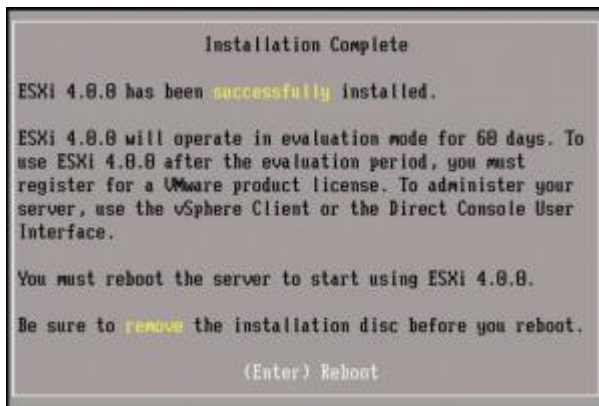
8. Installation in Progress

The installation process takes few minutes. While the ESXi is getting installed, it will display a progress bar as shown below.



9. ESXi Installation Complete

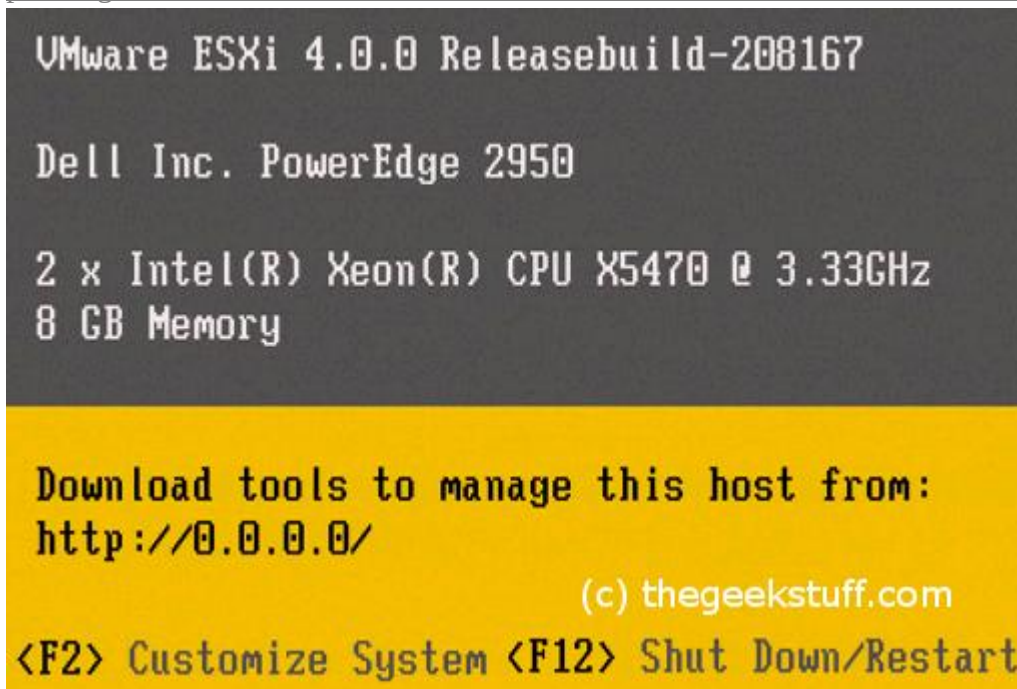
You will get the following installation completed message that will prompt you to reboot the server.



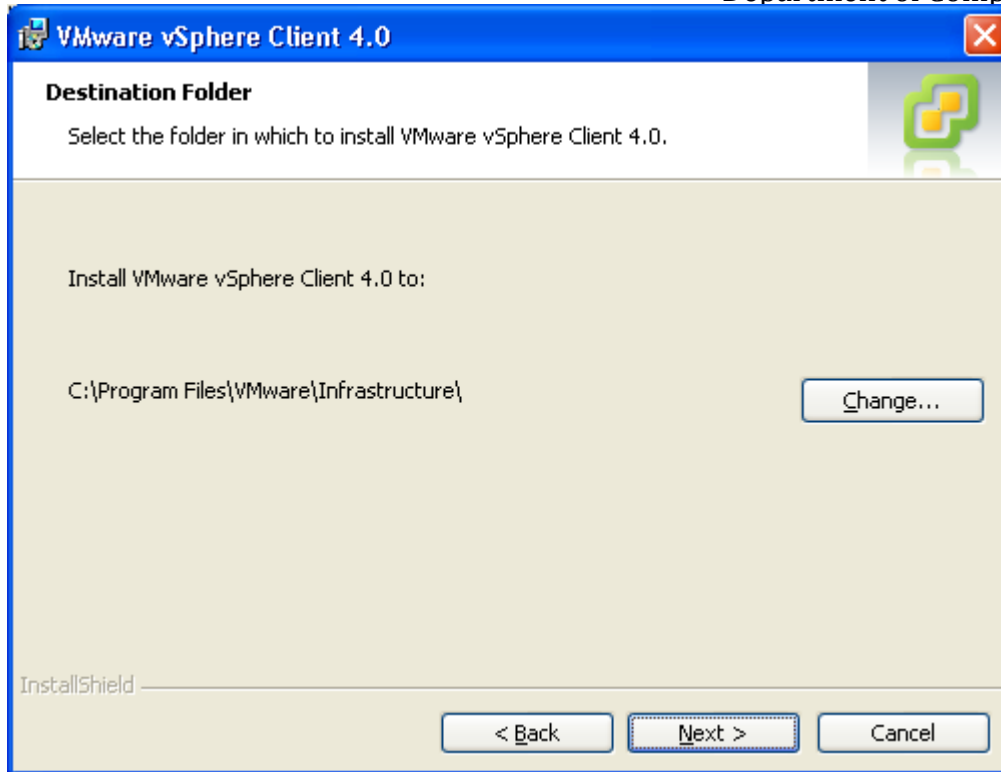
10. ESXi Initial Screen

After the ESXi is installed, you'll get the following screen where you can configure the system by pressing

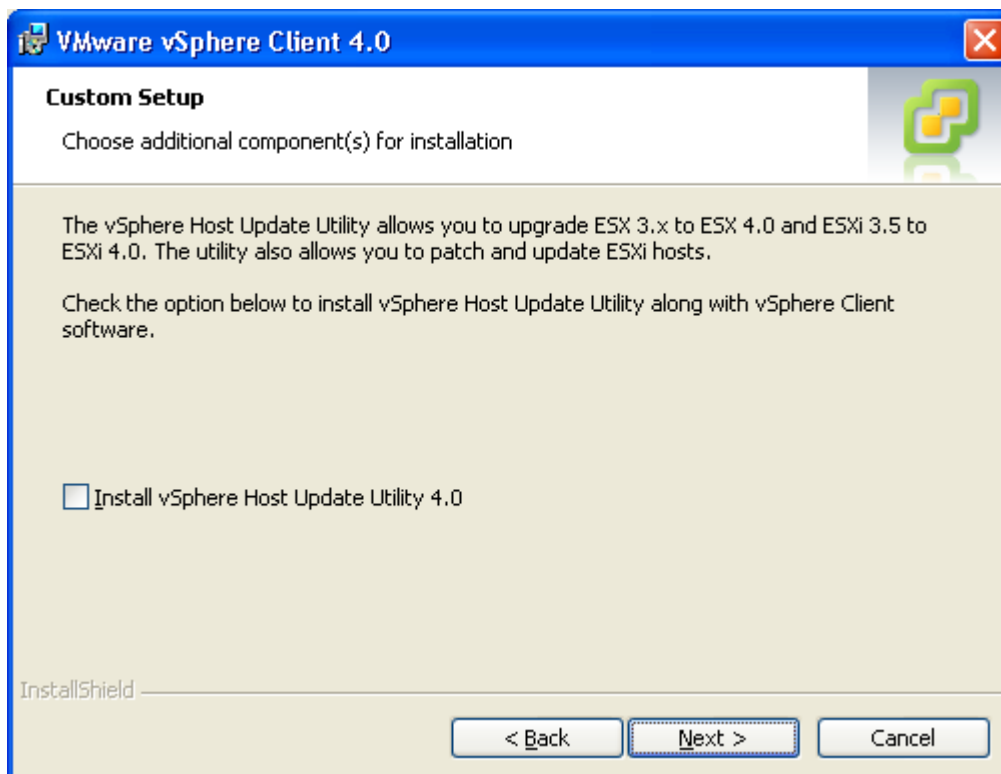
F2.



In the next article, let us review how to perform the initial ESXi configuration.



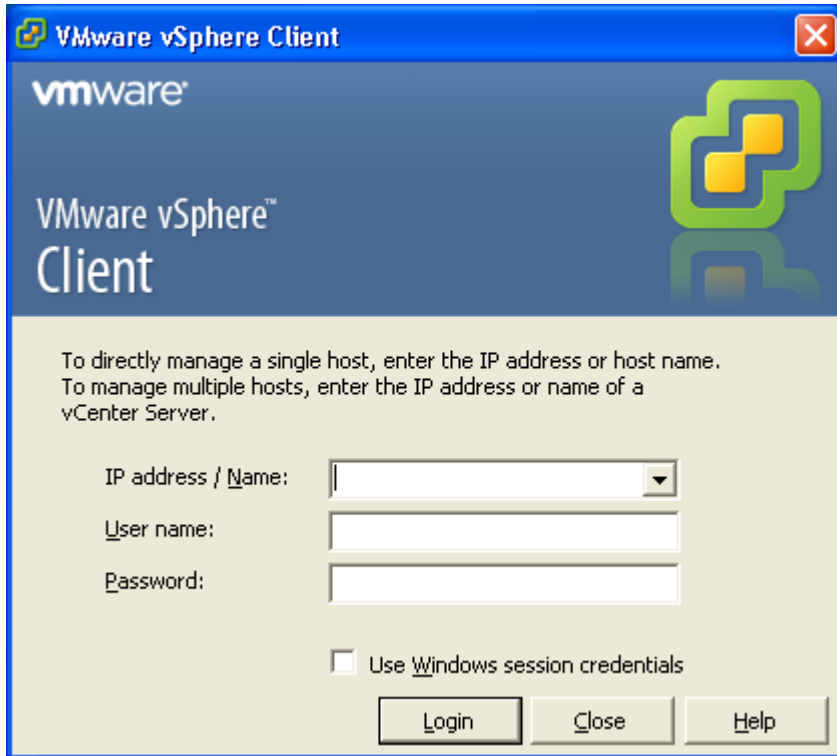
Since you are installing vSphere client to manage the newly installed ESXi 4.0 server, you don't need to select the 'Install vSphere Host Update Utility' check-box in the following screen.



Login to vSphere Client

After the installation, launch the vSphere client utility from your Windows host and provide the following information:

- ip-address of the VMware ESXi server.
- User name: root
- Password: root password for the ESXi server.



Ignore Security Warning

You might get a pop-up message about the security warning. Select the ‘Install this certificate and do not display any security warnings for {your-ip-address}’ check-box and click on ‘Ignore’.

VMware ESXi server is free. But, you should register at vmware website to get a license key. If you have not updated the ESXi server with the license key, you might also get the following “VMware Evaluation Notice” warning message.

VMware vSphere Client

You are now successfully logged-in to the vSphere client as shown below. In the upcoming articles, we’ll explain how to create new VM using vSphere client and how to manage a ESXi server using this utility.

Conclusion: - Thus we have installed OS on Virtual Machine Monitor

Experiment No: 7

Title: Offline/Online Migration of Virtual OS

Theory:

Migrating Virtual Machines

Migration is the process of moving a virtual machine from one host or storage location to another. Copying a virtual machine creates a new virtual machine. It is not a form of migration.

In vCenter Server, you have the following migration options:

- | | |
|--|---|
| Cold Migration | Moving a powered-off virtual machine to a new host. Optionally, you can relocate configuration and disk files to new storage locations. You can use cold migration to move virtual machines from one datacenter to another. |
| Migrating a Suspended Virtual Machine | Moving a suspended virtual machine to a new host. Optionally, you can relocate configuration and disk files to new storage location. You can migrate suspended virtual machines from one datacenter to another. |
| Migration with vMotion | Moving a powered-on virtual machine to a new host. Migration with vMotion allows you to move a virtual machine to a new host without any interruption in the availability of the virtual machine. You cannot use vMotion to move virtual machines from one datacenter to another. Some configurations include Metro vMotion, a feature that enables reliable migrations between hosts separated by high network round-trip latency times. Metro vMotion is automatically enabled when the appropriate license is installed. No user configuration is necessary. |
| Migration with Storage vMotion | Moving the virtual disks or configuration file of a powered-on virtual machine to a new datastore. Migration with Storage vMotion allows you to move a virtual machine's storage without any interruption in the availability of the virtual machine. |

Both migration of a suspended virtual machine and migration with vMotion are sometimes referred to as "hot migration", because they allow migration of a virtual machine without powering it off. Migration with vMotion is sometimes referred to as "live migration".

You can move virtual machines manually or set up a scheduled task to perform the cold migration.

Cloning a virtual machine or copying its disks and configuration file creates a new virtual machine. Cloning is not a form of migration.

Virtual Machine Configuration Requirements for vMotion